

Lab Title

Drawing 3D Cube and Pyramid using PyOpenGL

Objective

- Understand **basic 3D rendering**
 - Learn **OpenGL pipeline concepts**
 - Draw **cube and pyramid using vertices**
 - Understand **depth, projection, and transformations**
-

Software Requirement

Install required libraries in VS Code terminal:

```
pip install PyOpenGL PyOpenGL_accelerate pygame
```

Key Concepts (Simplified)

♦ 1. Coordinate System

- $X \rightarrow$ left/right
 - $Y \rightarrow$ up/down
 - $Z \rightarrow$ depth (into screen)
-

♦ 2. Important Functions

- `glBegin()` $\rightarrow$ Start drawing
 - `glVertex3f(x,y,z)` $\rightarrow$ Define point
 - `glColor3f(r,g,b)` $\rightarrow$ Set color
 - `glTranslatef(x,y,z)`
 - `glRotatef(angle, x, y, z)`
 - `glScalef(x,y,z)`
-

◆ 3. Depth Testing

`glEnable(GL_DEPTH_TEST)`

👉 Prevents back surfaces from overlapping front ones

PyOpenGL Code

```
from OpenGL.GL import *
from OpenGL.GLU import *
from OpenGL.GLUT import *
```

```
def init():
    glClearColor(0, 0, 0, 1)
    glEnable(GL_DEPTH_TEST)
```

```
def draw_cube():
    glBegin(GL_QUADS)
```

```
    # Top
```

```
    glColor3f(0,1,0)
    glVertex3f(1,1,-1)
    glVertex3f(-1,1,-1)
    glVertex3f(-1,1,1)
    glVertex3f(1,1,1)
```

```
    # Bottom
```

```
    glColor3f(1,0.5,0)
    glVertex3f(1,-1,1)
    glVertex3f(-1,-1,1)
    glVertex3f(-1,-1,-1)
    glVertex3f(1,-1,-1)
```

```
    # Front
```

```
    glColor3f(1,0,0)
    glVertex3f(1,1,1)
    glVertex3f(-1,1,1)
    glVertex3f(-1,-1,1)
    glVertex3f(1,-1,1)
```

```
    # Back
```

```
    glColor3f(1,1,0)
```

```
glVertex3f(1,-1,-1)
glVertex3f(-1,-1,-1)
glVertex3f(-1,1,-1)
glVertex3f(1,1,-1)
```

```
glEnd()
```

```
def draw_pyramid():
    glBegin(GL_TRIANGLES)
```

```
    # Front
    glColor3f(1,0,0)
    glVertex3f(0,1,0)
    glVertex3f(-1,-1,1)
    glVertex3f(1,-1,1)
```

```
    # Right
    glVertex3f(0,1,0)
    glVertex3f(1,-1,1)
    glVertex3f(1,-1,-1)
```

```
    # Back
    glVertex3f(0,1,0)
    glVertex3f(1,-1,-1)
    glVertex3f(-1,-1,-1)
```

```
    # Left
    glVertex3f(0,1,0)
    glVertex3f(-1,-1,-1)
    glVertex3f(-1,-1,1)
```

```
glEnd()
```

```
def display():
    glClear(GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT)
```

```
    # Cube
    glLoadIdentity()
    glTranslatef(1.5, 0, -7)
    draw_cube()
```

```
    # Pyramid
    glLoadIdentity()
    glTranslatef(-1.5, 0, -6)
```

```
draw_pyramid()

glutSwapBuffers()

def reshape(w, h):
    glViewport(0,0,w,h)
    glMatrixMode(GL_PROJECTION)
    glLoadIdentity()
    gluPerspective(45, w/h, 0.1, 100)

def main():
    glutInit()
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB | GLUT_DEPTH)
    glutInitWindowSize(640,480)
    glutCreateWindow(b"3D Shapes - PyOpenGL")

    init()
    glutDisplayFunc(display)
    glutReshapeFunc(reshape)

    glutMainLoop()

main()
```



Step-by-Step Explanation (Simple)

◆ Step 1: Initialize OpenGL

- Set background color
 - Enable depth testing
-

◆ Step 2: Draw Cube

- Uses **6 faces (quads)**
 - Each face = 4 vertices
-

◆ Step 3: Draw Pyramid

- Uses **4 triangles**

- Top point connects all sides
-

♦ Step 4: Transformation

`glTranslatef(x, y, z)`

👉 Moves object in 3D space

♦ Step 5: Perspective

`gluPerspective()`

👉 Makes objects look realistic (far = small)

🎯 Expected Output

- Right side → **Colored Cube**
 - Left side → **Colored Pyramid**
 - Objects appear in **3D perspective**
-

🧠 Common Student Mistakes

- Forgetting `GL_DEPTH_TEST`
 - Wrong vertex order (shape breaks)
 - Missing `glLoadIdentity()`
 - Objects overlapping incorrectly
-

📝 Practice Problems (For Student)

1.

Modify the program to:

👉 Change cube color dynamically

2.

Move pyramid **up and down**

3.

Add rotation:

`glRotatef(angle, 1,1,0)`

👉 Rotate cube continuously

4.

Change perspective:

👉 Try different `gluPerspective()` values

5.

Draw **3 objects**:

- Cube
 - Pyramid
 - Sphere (use `gluSphere()`)
-

6.

Create **keyboard control**:

- Press key → rotate object
 - Press key → move object
-

7.

Implement:

👉 **Scaling + Rotation + Translation together**

8.

Animate:

👉 Cube rotating + pyramid bouncing